

Projecting Future Energy Production from Operating Wind Farms in North America. Part I: Dynamical Downscaling

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ABSTRACT: New simulations at 12-km grid spacing with the Weather and Research Forecasting (WRF) Model nested in the MPI Earth System Model (ESM) are used to quantify possible changes in wind power generation potential as a result of global warming. Annual capacity factors (CF; measures of electrical power production) computed by applying a power curve to hourly wind speeds at wind turbine hub height from this simulation are also used to illustrate the pitfalls in seeking to infer changes in wind power generation directly from low-spatial-resolution and time-averaged ESM output. WRF-derived CF are evaluated using observed daily CF from operating wind farms. The spatial correlation coefficient between modeled and observed mean CF is 0.65, and the root-mean-square error is 5.4 percentage points. Output from the MPI-WRF Model chain also captures some of the seasonal variability and the probability distribution of daily CF at operating wind farms. Projections of mean annual CF (CF_A) indicate no change to 2050 in the southern Great Plains and Northeast. Interannual variability of CF_A increases in the Midwest, and CF_A declines by up to 2 percentage points in the northern Great Plains. The probability of wind droughts (extended periods with anomalously low production) and wind bonus periods (high production) remains unchanged over most of the eastern United States. The probability of wind bonus periods exhibits some evidence of higher values over the Midwest in the 2040s, whereas the converse is true over the northern Great Plains.

SIGNIFICANCE STATEMENT: Wind energy is playing an increasingly important role in low-carbon-emission electricity generation. It is a “weather dependent” renewable energy source, and thus changes in the global atmosphere may cause changes in regional wind power production (PP) potential. We use PP data from operating wind farms to demonstrate that regional simulations exhibit skill in capturing actual power production. Projections to the middle of this century indicate that over most of North America east of the Rocky Mountains annual expected PP is largely unchanged, as is the probability of extended periods of anomalously high or low production. Any small declines in annual PP are of much smaller magnitude than changes due to technological innovation over the last two decades.

KEYWORDS: Interannual variability; Interdecadal variability; Numerical analysis/modeling; Regional models; Subseasonal variability; Trends

1. Introduction

a. Motivation

Wind energy is an increasingly important source of renewable and low-carbon emission electricity worldwide and in the United States. Global wind energy installed capacity (IC) grew at an annualized rate of over 14% between 2006 and 2020, and, by the end of 2020, 742 GW of wind energy IC was generating approximately 6.5% of global electricity supply (Global Wind Energy Council 2021). The Global Wind Energy Council anticipates additional IC increases of nearly 94 GW per year until 2025 (Global Wind Energy Council 2021). Wind energy IC expansion to 2050 could reduce greenhouse gas emissions and the associated global mean temperature rise at 2100 by 0.3°–0.8°C depending on the precise radiative forcing applied and wind energy expansion scenario followed (Barthelmie and Pryor 2021).

Wind energy represents 29% of total U.S. electricity generation capacity additions over the last decade (Wiser et al. 2021).

At the end of 2020 the total U.S. IC was 122 GW (Wiser et al. 2021), rising to 135 GW at the end of 2021 (American Clean Power 2022). According to the U.S. Energy Information Administration (EIA), in 2020, 3.7 trillion kW h of electricity was sold to end-use customers (<https://www.eia.gov/energyexplained/electricity/electricity-in-the-us.php>), 8.4% of which came from wind turbines (<https://www.eia.gov/energyexplained/electricity/how-electricity-is-generated.php>).

The levelized cost of energy (LCoE) from wind turbines, expressed as cost per megawatt hour (MW h) of electricity generated has fallen precipitously (Bistline and Young 2019; Dong et al. 2021). LCoE for wind turbines deployed onshore in the United States is now approximately USD 40 per MW h (Beiter et al. 2021). Forecasts of LCoE from onshore wind in the United States in 2050 are generally between USD 20 and 30 per MW h, whereas those for offshore wind are USD 40–60 per MW h (Beiter et al. 2021).

Wind energy is also a weather-dependent source of electricity generation. This has prompted questions about the potential for greenhouse gas induced climate change to alter the spatial patterns, seasonality, and/or magnitude of wind resources (Pryor et al. 2020d). Detection and attribution of

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historical changes, including secular trends, and development of robust projections of future changes in regional wind resources and potential electrical power production is rendered complex by three primary factors that are described in the following paragraphs.

First, wind resources in different locations are dictated by atmospheric processes acting across multiple scales (Dörenkämper et al. 2020; Hahmann et al. 2020; Haupt et al. 2019). At some locations, local thermo-topographic flow and orographic channeling are critical (Barthelmie et al. 2016), whereas at other locations the flow regime is more strongly influenced by meteorological phenomena manifest at the mesoscale (e.g., low-level jets; Aird et al. 2021; Barthelmie et al. 2020) or the passage of transitory midlatitude cyclones (Pryor et al. 2012a). Flow across these scales and their driving phenomena may exhibit different and regionally specific responses to greenhouse gas induced changes in the global atmosphere. Most regions with high wind resources are located in the midlatitudes close to the primary storm tracks along which major synoptic-scale midlatitude cyclones move. Even in the case of locations where the wind resource is dominated by the presence of transitory midlatitude cyclones, the presence, magnitude and even sign of expected change is not easy to postulate. The dynamics (e.g., intensity, as measured, e.g., by pressure gradients) and tracking of these extratropical cyclones (ETCs) exhibit sensitivity to changes in atmospheric water vapor and baroclinicity, and the net impact of changes in these parameters on the kinetic energy content of eddies (e.g., ETC) is likely to be highly regionally specific (Lehmann et al. 2014; Wang et al. 2017). Enhanced greenhouse gas concentrations have the potential to lead to so-called Arctic amplification (i.e., relative warming of the polar regions), which may cause changes in the preferred location and/or variability in the polar jet stream and hence these storm tracks (Harvey et al. 2020; Stendel et al. 2021). There may also be offsetting impacts on the geographic anchoring of midlatitude storm tracks from processes acting on shortwave and longwave radiation (Shaw et al. 2016).

Second, very few observational datasets of wind speed at wind turbine hub height or wind turbine power production are publicly available to build statistical models or evaluate dynamical models (Dörenkämper et al. 2020; Hahmann et al. 2020; Kusiak 2016; Pryor and Hahmann 2019). Past research has demonstrated divergent trends in different observational records of wind speeds at 10 m above ground level (AGL) (Pryor et al. 2009) and that interannual variability in such records is a poor proxy for interannual variability in actual wind power generation (Millstein et al. 2022). Because wind power density (WPD) in the flow scales with the cube of the wind speed and the relationship between the wind speed and the electrical power production by a wind turbine is highly nonlinear (see Fig. 1a) (Pryor et al. 2020d), fidelity with respect to annual mean wind speed at 10 m AGL cannot be used to infer fidelity with respect to the property of interest (WPD or electrical power generation). The mean wind speed at any site is not a robust or resilient estimator of the central tendency of the probability distribution of wind speeds (U) because they do not fit a Gaussian distribution but tend to

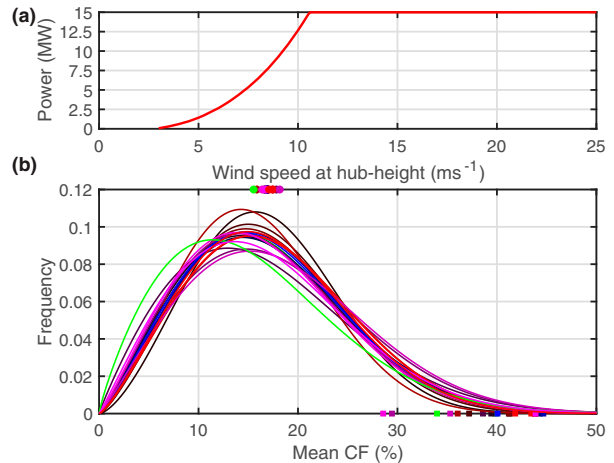


FIG. 1. (a) Illustrative power curve (showing the relationship between the wind speed at hub height and the amount of electrical power produced by a wind turbine. This example is for the IEA 15 MW reference wind turbine [details are in Gaertner et al. (2020)]. (b) Two-parameter Weibull distributions fitted to hourly wind speeds at hub height for 2010–49 from WRF for each of the wind farm locations considered here. The line coloring denotes the region in which the wind farm is located (see Fig. 3, below). The filled circles on the top axis denote the mean wind speed computed across the entire 40-yr simulation, and the filled squares on the bottom axis show the mean capacity factors.

more closely approximate a Weibull distribution (Pryor et al. 2005), wherein the cumulative probability distribution is given by

$$P(U) = 1 - \exp\left[-\left(\frac{U}{c}\right)^k\right], \quad (1)$$

where k and c = shape and scale parameters.

Third, there is substantial variability in wind climates that is due to the action of important internally forced climate variability across a range of time scales (Jung et al. 2019; Lledó et al. 2018; Pryor and Hahmann 2019; Pryor et al. 2018; Schoof and Pryor 2014; Zeng et al. 2019). For example, wind climates and resources over North America exhibit region-specific responses to three dominant internal climate modes (Schoof and Pryor 2014). The Pacific–North American index (PNA), El Niño–Southern Oscillation (ENSO) and Pacific decadal oscillation (PDO) dominate low-frequency variability in the west (Yu et al. 2015). Generally, El Niño is associated with lower wind speeds and the La Niña phase with higher wind resources (Yu et al. 2015). Over the central plains, ENSO, PNA, and an index representing the Northern Annular Mode (NAM) or the Arctic Oscillation (AO) play more dominant roles (Coburn 2021; Klink 2007). Over the Northeast, either NAM or the North Atlantic Oscillation are dominant (Pryor and Ledolter 2010). Analyses have shown that the recent increases in capacity factors from operating wind farms in North America 2010–17 are largely attributable to more favorable wind climate resulting from a transition toward negative value of the tropical northern Atlantic index

that reflects sea surface temperature anomalies in the North Atlantic Ocean (Zeng et al. 2019).

b. Past research

Three primary approaches have been used in previous research focused on quantifying possible future changes in wind resources and electricity generation (Pryor et al. 2020d): 1) Analyses of direct wind speed output from global Earth system models (ESMs; Carvalho et al. 2021; Kulkarni and Huang 2014). 2) Analyses of output from dynamical downscaling simulations with regional models or regionally refined global models (Pryor and Barthelmie 2011; Pryor et al. 2012b; Wang et al. 2018). 3) Application of statistical downscaling using predictors output from regional and/or global models (Pryor and Barthelmie 2014; Pryor et al. 2005; Reyers et al. 2015). Projected variability and change in wind climates exhibit a strong dependence on the (i) downscaling approach used (numerical v statistical), (ii) ESM used to provide the lateral boundary conditions (LBC) and/or predictors, (iii) regional model applied, (iv) greenhouse gas forcing (Shared Socio-economic Pathway scenario), and (v) time period considered (Kjellström et al. 2018; Pryor and Schoof 2010; Pryor et al. 2012b). Nevertheless, some convergence with regard to potential changes in regional wind resources is beginning to emerge (Fig. 2).

c. Objectives

This two-part series of papers reports robust wind power projections for future decades over North America. We sample across analysis tools: regional modeling in Part I and statistical downscaling in Coburn and Pryor (2023, hereinafter Part II); in Part II, we also sample the uncertainty associated with use of predictors from different ESM and different climate forcing scenarios.

A key novelty of this research is that our analyses use actual daily *expected power* from operating wind farms across North America to both evaluate the numerical simulations (Part I) and build statistical models (Part II). Expected power is the electrical power that would be produced at a given windfarm if (i) no operations and maintenance (O&M) activities occur (Wiser et al. 2021). Most estimates suggest wind turbine availability for onshore wind projects is greater than 97%; thus O&M activities reduce power production by less than 3% (Artigao et al. 2018). (ii) The wind farm is not subject to curtailment for grid management (less than 2% of available power in the United States during 2018; Zhang et al. 2020). (iii) No electrical power losses occur (less than 2%; Staffell and Green 2014). Expected power relies on wind speeds measured within the wind farm and thus these values do include power losses due to the influence of wind turbine wakes. Such losses are expected to be less than 5% of available power for the typical dimensions of U.S. onshore wind farms (Staid et al. 2018), although use of a wider range of possible wake losses (1%–10%) has been proposed for use in wind farm layout optimization studies (Clifton et al. 2016).

For reasons of data confidentiality, the expected power in MWh per day from each wind farm is converted to capacity

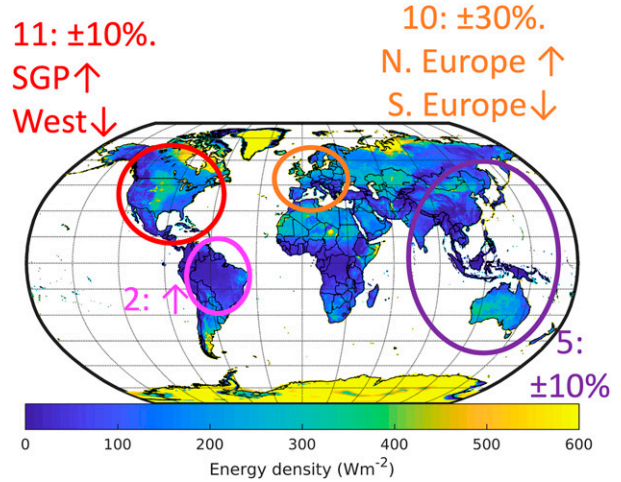


FIG. 2. Subjective precis of wind resource projections in different regions summarized from the review in Pryor et al. (2020d). The information is provided in terms of the number of studies summarized for each given region (number before the colon) and then a precis of the findings. The value $\pm x\%$ indicates the inferred range of changes in mean wind power density at the end of the current century vs the historical climate. The arrows indicate the sign of difference indicated by the majority of studies in each subregion. Fewer studies have been performed for South Asia/Australia, and there is no consensus on the direction of change; therefore, no arrow is shown. The background shading shows WPD computed here using wind speeds at a height of 100 m from hourly output for 1979–2018 from the ERA5 reanalysis (Hersbach et al. 2020).

factors (CF) prior to presentation herein. CF are the amount of electrical power produced (in this case, the expected power) divided by the potential power produced if all wind turbines at a given facility run at their rated capacity. CF values presented herein are not equivalent to *net* CF from operating wind farms, due to power-induced losses from factors i–iii described above (Lee and Fields 2021) but do reflect the influence of variability in the geophysical wind resource that may be impacted by global climate change. Access to these operational data represents a unique opportunity to evaluate fidelity in the numerical simulations in a context that is highly relevant to the renewable energy industry. These CF are computed from expected power scale with actual power production (net CF) under normal wind farm operation and are directly related to financial revenue and the LCoE via (Beiter et al. 2021)

$$\text{LCoE} = \frac{C_0 \times \text{CRF} + \text{O\&M}}{\text{nhf} \times \text{NCF}}, \quad (2)$$

where C_0 = cost of capital ($\$ \text{ kW}^{-1}$), O&M = operations and maintenance expenditure [$\$ (\text{ kW yr})^{-1}$], CRF = capital recovery factor that converts capital costs to annuity ($\% \text{ yr}^{-1}$ accounting for cost of finance), nhf = number of hours in a year, and NCF = net capacity factor (%). This simple model illustrates that there is an inverse linear relationship between LCoE and CF.

The research described in this two-part series of articles is designed to address four key research objectives:

- 1) illustrate the limitations inherent in studies that draw inferences about possible changes in wind power production based solely on near-surface wind speeds from ESM,
- 2) provide the first evaluation of CF derived from moderate-resolution (12-km grid spacing) WRF output and wind turbine power curves versus CF derived from observed expected power,
- 3) develop robust projections of annual mean CF (CF_A) and 50th-percentile annual mean CF [$P50(CF_A)$] at sites across North America from WRF output and a broad ensemble of statistically downscaled ESM from phase 6 of the Coupled Model Intercomparison Project (CMIP6) (Part II), and
- 4) quantify possible changes in the probability of anomalously low production (wind droughts) and anomalously high production (wind bonus) periods.

Convergence in the sign, magnitude, and spatial manifestations of changes in wind conditions due to global climate non-stationarity derived from the two independent regionalization approaches and across multiple ESMs can be used to infer higher confidence or credibility in those projections. The authors are not aware of any previous study that has employed actual wind power production from operating wind farms to evaluate climate-scale WRF simulations and/or generate statistically downscaled wind power production estimates.

2. Methods

a. Expected power

Time series of daily expected power from operating wind farms are supplied by a wind farm owner-operator on the condition of anonymity and thus are anonymized as follows:

- 1) They are referred to only in terms of the region in which they are located. The dataset comprises expected power from seven wind farms operating in the northeastern United States and eastern Canada (referred to herein as Northeast), three in the Midwest (Midwest), seven in the southern Great Plains (SGP), and two in the northern Great Plains (NGP) (Fig. 3).
- 2) They are converted into estimated CF as follows:
 - (i) Any daily value with expected power below 0 is replaced with NaN. A total of 324 values below 0 were found in the entire data record for all sites.
 - (ii) Any daily value with expected power in megawatt hours greater than 24 times the installed capacity is set to 24 times the installed capacity.
 - (iii) The sum of daily values of expected power are divided by the installed capacity at each site to generate daily CF.

The duration of the daily expected power time series from each of the operating wind farms varies from location to location and ranges from a minimum of one year to a maximum of six years.

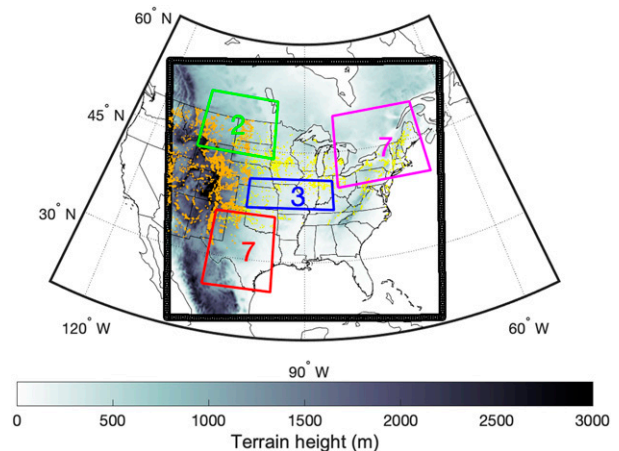


FIG. 3. Map of the WRF simulation domain (and terrain height) and outlines of the regions (colors), with the number of wind farm assets in each region, for which expected power estimates are presented. Also shown are the locations with CF estimates from the NREL Wind Prospector dataset that fall within the simulation domain. Locations with values from the WP:EWD are shown in yellow, and those from the WP:WWD are shown in orange.

b. WRF simulations

Numerical simulations are performed using the Weather Research and Forecasting (WRF) Model, version 3.8.1. WRF has been widely applied and validated in regional climate and wind energy applications (Bukovsky and Mearns 2020; Hahmann et al. 2020; Pryor et al. 2021). Here WRF is applied with a grid spacing of 12 km $\times$ 12 km, a time step of 60 s, and 41 vertical levels for a domain centered over North America (Fig. 3). Although decreasing grid spacing does not always increase model fidelity with respect to flow fields (Draxl et al. 2014a; Mass et al. 2002), a horizontal grid spacing (Δx) $\sim$ 4 km is more typical of WRF simulations performed for wind resource assessments (i.e., in the development of so-called wind atlases; Draxl et al. 2014a; Hahmann et al. 2021). A recent evaluation of surface wind speeds from WRF simulation in the complex terrain of Norway found “lower error when the grid spacing is reduced from 27 km via 9 to 3 km. From 3 to 1 km, the error is not further reduced” (Solbakken et al. 2021). Use of $\Delta x \sim$ 4 km is consistent with recommendations that Δx should be greater than the depth of the planetary boundary layer to avoid development of unrealistic flow features (Rai et al. 2019). Use here of a 12-km grid spacing means that not all of the factors that dictate local wind resources can be captured (Dörenkämper et al. 2020; Haupt et al. 2019). However, setting $\Delta x = 12$ km also renders very long simulations over a large domain computationally feasible. We further postulate that in regions with a very dynamic synoptic climate (such as much of the eastern two-thirds of North America) much of the climate change signal is likely to be due to changes in the tracking, frequency, morphology, and intensity of mid-latitude transitory cyclones (Dai and Nie 2022; Harvey et al. 2020; Sinclair et al. 2020). It will thus be manifest at scales represented in simulations performed using $\Delta x = 12$ km. We

further note that previous research that has sought to examine possible impacts from nonstationarity on wind resources has generally adopted lower spatial resolution (see summary in Pryor et al. 2020d) and/or has employed higher spatial resolution over a considerably smaller domain (Losada Carreño et al. 2020).

Lateral boundary conditions are drawn from the ensemble member (r1i1p1) of the Max Planck Institute Earth System Model, low resolution (MPI-ESM-LR; Giorgetta et al. 2013), for mid-2009 to 2049 under the RCP 8.5 $W\ m^{-2}$ forcing. MPI-ESM-LR is selected from the CMIP5 suite because it has a transient climate response of $2^{\circ}C$ and equilibrium climate sensitivity of $3.6^{\circ}C$ (ECS) that are typical of the CMIP5 multimodel mean (Bukovsky and Mearns 2020). MPI-ESM-LR exhibits above average skill in the representation of major climate modes, including NAM (Gillett and Fyfe 2013), AO (Jin-Qing et al. 2013), and ENSO (Bellenger et al. 2014) that play key roles in inducing low-frequency variability in wind climates over North America. MPI-ESM-LR also exhibits relatively high fidelity for Northern Hemisphere midlatitude planetary and baroclinic wave energy (Di Biagio et al. 2014), and aspects of Northern Hemisphere cyclogenesis (Seiler and Zwiers 2016).

The WRF configuration employed here uses physics settings selected based on previous research. Extensive work with the New European Wind Atlas (NEWA) project (Hahmann et al. 2020) and other previous work found very weak sensitivity of wind fields to the shortwave and longwave radiation schemes and to the cumulus and microphysics schemes. Thus, the following computationally efficient schemes are selected. Rapid Radiative Transfer Model for longwave radiation (Mlawer et al. 1997), Dudhia for shortwave radiation (Dudhia 1989), the Kain–Fritsch cumulus scheme (Kain 2004), and Eta microphysics scheme (Ferrier et al. 2002). Wind speeds and wind profiles show a much stronger sensitivity to planetary boundary layer (PBL) and surface schemes (Draxl et al. 2014b; Gómez-Navarro et al. 2015; Hahmann et al. 2020; Qian et al. 2003; Yang et al. 2019). Noting there are some constraints related to scheme compatibility. Here we use the Mellor–Yamada–Nakanishi–Niino 2.5 PBL scheme (Nakanishi and Niino 2006), revised MM5 Monin–Obukhov scheme (Jiménez et al. 2012), and the Noah land surface model (Tewari et al. 2004). This combination was also included in the NEWA sensitivity analysis (Hahmann et al. 2020). The combination of PBL and surface physics parameterizations has an impact on the simulated wind speed and turbulence fields to a degree that is location specific because of the dependence of near-surface flow on orographic complexity (Lorente-Plazas et al. 2016), the surface energy balance and stability conditions (Hahmann et al. 2020; Lorente-Plazas et al. 2016), and surface roughness (Hahmann et al. 2020). For this reason, the optimal combination of schemes to minimize model error relative to measurements of wind speed on meteorological masts also appears to be dependent on the environment under consideration. For example, an analysis of WRF configurations against a month of measurements from a tall mast in Denmark found all seven PBL schemes tested underestimated wind speeds close to wind turbine heights during the nighttime hours and overestimated them during the daytime (Draxl et al. 2014b). That analysis further found the Mellor–Yamada–Janjić

PBL scheme performed best in stable and very stable conditions, the Asymmetric Convective Model, version 2, PBL scheme exhibited lowest bias and smallest root-mean-square error in near-stable and neutral conditions and the Yonsei University compared best to observations during unstable conditions (Draxl et al. 2014b). A further analysis of WRF simulated wind turbine hub-height wind speeds found results from the MYNN PBL scheme generally agreed better with observations than simulations using the YSU PBL scheme under both weak and strong wind conditions (Yang et al. 2019). The MYNN PBL scheme is widely used and is selected for this work as a compromise between lower-order closure schemes that are expected to exhibit lower fidelity and more computationally demanding and less well-characterized higher-order or nonlocal closure schemes. Default values are used for the 14 coefficients in the MM5 surface scheme due to past research that has shown near-surface wind speeds exhibit only a weak dependence on the specific values selected (Yang et al. 2017).

The modified MODIS International Geosphere–Biosphere Programme land-cover classification dataset (20-category vegetation) (Friedl et al. 2010) is used to supply land surface characteristics. The default WRF lookup tables for surface roughness length z_0 are used here, but we note that analyses performed within the NEWA project suggested that the z_0 for forests may be underestimated (Hahmann et al. 2020). Sea surface temperature (SST) values are updated every 24 h, derived from the MPI-LR ocean model output. The SST data are remapped using a bilinear interpolation function from the ocean model grid (122×101) to a global cylindrical equidistant grid to match the atmospheric model grid (256×220). A weighted-average interpolation is used in WRF preprocessing system to generate more-realistic SST values around coastlines in the model domain.

The model is cold restarted each month to prevent model drift from running a multiyear simulation (Pan et al. 1999). The decision to use cold restarts and a 6-h spinup period ensures that the WRF simulation closely follows the driving ESM (MPI-ESM-LR) and thus makes the dynamical downscaling more directly comparable to the statistical downscaling described in Part II of this analysis. Monthly cold restarts circumvent the lack of mass conservation in WRF but prevent slowly evolving quantities (e.g., soil moisture) being entirely dictated by the regional model (Caldwell et al. 2009; Qian et al. 2003). A further practical advantage of this approach is that simulations of multiple months can be run in parallel (Losada Carreño et al. 2020). The 3D fields of all key atmospheric variables are output once per hour.

c. NREL Wind Prospector

The NREL Wind Prospector: Eastern Wind Dataset (WP:EWD) and the Wind Prospector: Western Wind Dataset (WP:WWD) (3TIER 2010; Brower 2009; Pennock 2012) contains net CF estimates for a nominal hub height of 80 m AGL at potential wind farm locations across the eastern and western United States, respectively (Fig. 3). These estimates are based on WRF simulations of 2004–06 using a 2-km grid

spacing. WP:EWD and WP:WWD output within grid boxes of varying dimensions centered on each operating wind farm considered here are also compared with the observationally derived estimates from expected power and the new MPI-WRF simulation. Note that, as shown in Fig. 3, the WP:EWD and WP:WWD output undersamples the southern Great Plains and does not include locations in Canada.

d. Analysis methods

The specific power curve for wind turbines operating at each site is applied to hourly wind speed time series of wind speeds at the hub height to determine the hourly power output, P_h . The values are summed, and the result is divided by the number of hours (nhr) in each calendar year (e.g., 2010, 2011, 2029, or 2037) multiplied by the rated capacity of the wind turbine (RC) to obtain the annual mean capacity factors (CF_A):

$$CF_A = \frac{\sum_{h=1}^{nhr} P_h}{nhr \times RC}. \quad (3)$$

In the evaluation, the first 6 years of the WRF-MPI simulation output (i.e., nominal years of 2010–15) are taken as representative of the contemporary climate and are compared with the mean CF ($\langle CF \rangle$) derived from expected power. Because expected power varies seasonally, only full years of data are used to determine $\langle CF \rangle$ used in the model evaluation. At the two wind farms where the precise wind turbine power curves are not available, a scaled version of a power curve for a similar wind turbine is used (Pryor et al. 2020b).

A nonparametric Mann–Kendall test is applied to the time series of CF_A at each wind farm to examine the presence (or absence) of secular trends (Wilks 2011). Analyses of low-frequency variability in wind power projections presented here employs metrics widely used in the wind energy industry to quantify interannual variability (IAV) of expected annual energy production (AEP) from proposed wind farms for project financing (Pryor et al. 2018). Although IAV(AEP) is a function of the record duration, precise time period covered and location, the standard deviation of AEP is often approximated as 6% of the mean value (Brower 2012; Pullinger et al. 2017; Raftery et al. 1998). A prior model-based analysis over eastern North America found that the interquartile range (IQR: 75th–25th percentile) extended ± 3 percentage points around the 50th-percentile AEP (Pryor et al. 2018). $P50(AEP)$ is the AEP projected to be equaled or exceeded on 50% of years during wind farm operation, and $P90(AEP)$ is the AEP that is associated with a 10% risk of not being reached (Pryor et al. 2018). To be consistent with the data anonymization, we present the $P50(CF_A)$ and $P90(CF_A)$ in each decade. We use bootstrap resampling with replacement (Wilks 2011) of CF_A in the 2010–19 output to derive confidence intervals on $P50(CF_A)$ for the contemporary climate and evaluate any differences between $P50(CF_A)$ in each subsequent decade relative to 2010–19 using those confidence intervals. If $P50(CF_A)$ for 2040–49 lies beyond the 95% confidence intervals on

$P50(CF_A)$ in the 2010–19 period, the difference is deemed statistically significant.

The probability of regionally coherent extended periods with anomalously low (or high) electricity production are of particular interest in terms of grid integration at the macro-scale and for company financial health at the wind plant level (Pryor et al. 2020a; Rinaldi et al. 2021). Observed and modeled CF exhibit clear seasonality (Pryor et al. 2020a), thus there is a need to examine periods of anomalously high or low CF in a seasonally adjusted context. Accordingly, projections of the probability of “wind droughts” and “wind bonus” describe 30-day periods of anomalously low or high power production for that time of the year. The method used to identify wind drought and wind bonus periods from the WRF output is as follows. First, the daily mean CF at the given wind farm is computed for the entire period of output. The values for 15 January 2010 to 15 December 2049 are used to compute a 30-day running-mean CF centered on each calendar date. The 20th- and 80th-percentile 30-day running-mean CF for that day of the year are found. For example, for 4 July, the sample of 40 years of 30-day running-mean CF centered on that date are used to find the 20th percentile (i.e., the 8th-smallest value) and the 80th percentile (i.e., the 8th-largest value in the 40 years). Each 30-day running-mean CF is then evaluated to determine whether it falls in the lowest 20% or the upper 20% on that calendar date and thus can be characterized as a wind drought or wind bonus period, respectively. If there is a tendency toward an increased frequency of wind droughts, then the probability that any period drawn at random will fall in the lower 20% of values will be higher in later decades than early in the simulation. These wind droughts and wind bonus thresholds are site specific. Thus, we are addressing the following questions: Will periods with extended, relatively low productivity at a given location change? Will periods with extended, relatively high productivity at a given location change?

Many studies have analyzed once ESM daily wind speed output at 10 m at typical ESM resolutions of approximately 100–200 km $\times$ 100–200 km, to derive information about possible changes in annual mean wind speed or WPD and thus infer information about likely wind power production. Prior to analysis of the WRF output to address potential changes in wind power production, we use it to illustrate that direct use of ESM wind speed output (i.e., without downscaling), irrespective of the precise methodology applied, does not yield results appropriate for use by the wind energy industry. These analyses examine the relationship between CF_A and metrics derived by degrading the WRF output to represent variables commonly used in ESM direct wind power production analyses:

Metric 1: Annual mean wind speed at hub height ($\langle WS_{HH} \rangle$ 12 km) is computed using once hourly wind speeds.

Metric 2: Mean spatially averaged wind speed at hub height [$\langle WS_{HH} \rangle$ (9 by 9)] in each year is computed by first spatially averaging the hourly output to the typical resolution of an ESM (108 km $\times$ 108 km), that is, over 9 by 9 grid cells centered on each wind farm.

Metric 3: Annual mean wind speeds at 10 m [$\langle WS_{10m} \rangle$ (12 km)] are computed using the once hourly wind speeds at 10 m (i.e., height that ESM wind speed output is typically archived).

Metric 4: Annual mean wind speeds at 10 m over 9 by 9 grid cells centered on the wind farm [$\langle WS_{10m} \rangle$ (9 by 9)] are computed by first spatially averaging the hourly wind speeds.

Metric 5: Wind power density at wind turbine hub height [$\langle WPD_{HH_from_10m} \rangle$ (9 by 9)] is computed by spatially averaging 10-m wind speeds and then “lifting” them to hub height (HH) using a power law:

$$U = WS_{10m} \left(\frac{HH}{10} \right)^{1/7} \quad (4)$$

The resulting value for U is converted to WPD in each hour using

$$WPD = \frac{1}{2} \rho U^3 \quad (5)$$

where ρ = air density (1.225 kg m^{-3}). Note the 1/7 exponent has been found to generate a wind speed profile with too little vertical shear over the Great Plains (Barthelmie et al. 2020) but is commonly used in the vertical extrapolation performed within the literature.

Additional analyses are also performed to assess the impact of temporal *and* spatial averaging:

Metric 6: CF_A daily WS_{HH} (12 km) is computed from the daily mean wind speed at HH and Eq. (3).

Metric 7: CF_A daily WS_{HH} (9 by 9) is computed by spatially averaging hourly HH wind speeds over 9 by 9 grid cells and then averaging in time, applying the wind turbine power curve and Eq. (3).

Metric 8: CF_A daily WS_{10m} w/power law (9 by 9) is calculated from the daily mean spatially averaged 10-m wind speed followed by application of the power law with an exponent of 1/7 to lift to HH [following Eq. (5)]. The wind turbine power curve is applied to the resulting wind speed, followed by Eq. (3).

3. Results

a. A cautionary note on the use of near-surface wind speeds from Earth system models for assessing possible changes in wind power generation

Analyses of WRF output degraded to mimic that typically available from global model archives are shown in Fig. 4, and are briefly summarized below:

Metric 1: There is a relatively strong linear relationship between the modeled CF_A and $\langle WS_{HH} \rangle$ (12 km) at each individual location (Fig. 4a). However, the slope and intercept of the relationship varies across

sites. While the sites considered herein have a relatively narrow range of mean wind speeds at wind turbine hub height, the mean CF_A s computed across all 40 years of the simulation span a much greater range (Fig. 1b). This is due to variations in the probability distribution and the nonlinearity inherent in wind turbine power curves (Fig. 1a). The range of $\langle WS_{HH} \rangle$ (12 km) across the 19 locations is $7.7\text{--}9.1 \text{ m s}^{-1}$ (a variation of $\sim \pm 8\%$ around the mean), and the range of mean modeled CF extends from 28.5% to 44.7% (a variation of $\sim \pm 20\%$ around the mean value sampled across all sites) (Fig. 1b).

Metric 2: The relationship between $\langle WS_{HH} \rangle$ (9 by 9) and CF_A exhibits more year-to-year variability at individual locations and is less consistent across space (Fig. 4b). This illustrates that spatial resolution above that typical of ESM is important to capturing the relationship between wind speeds and CF_A .

Metric 3: The relationship between $\langle WS_{10m} \rangle$ (12 km) and CF_A exhibits more year-to-year variability at individual locations and is even less consistent across space (Fig. 4c). This is likely because of greater spatial and temporal variability in wind speeds at 10-m height.

Metric 4: The relationship between $\langle WS_{10m} \rangle$ (9 by 9) and CF_A at an individual wind farm and across space is less consistent still, reemphasizing the importance of model spatial resolution (Fig. 4d).

Metric 5: $\langle WPD_{HH_from_10m} \rangle$ (9 by 9) exhibits an even weaker relationship with CF_A at a given location and is even less consistent across space (Fig. 4e).

Metric 6: CF_A and CF_A daily WS_{HH} (12 km) are highly correlated, but the values derived from the daily mean wind speed are systematically negatively (i.e., low) biased (Fig. 4e). Averaged across all 40 years at each location and then across all 19 locations, the mean difference in $\langle CF_A \rangle$ and CF_A daily WS_{HH} (12 km) is +1.7 percentage points.

Metric 7: CF_A daily WS_{HH} (9 by 9) values again deviate markedly from CF_A , although at an individual site the two estimates are highly correlated (Fig. 4g). There is no systematic bias in CF_A from the averaged output due to the proximity of some locations to low-roughness water bodies, which means the spatial averaging compensates for the temporal averaging.

Metric 8: CF_A daily WS_{10m} w/power law (9 by 9) exhibits substantial negative bias relative to CF_A (Fig. 4h).

These analyses imply that models operating at low spatial resolution cannot capture key scales of motion relevant to wind resources and electrical power production. Further, there is no consistent generalizable relationship between likely electricity production and wind speeds or WPD sampled at spatial resolutions representative of ESM. A difference in annual mean wind speed at hub height in a given location need not be associated with the same fractional change in CF_A or indeed any change in CF_A . Additionally, a difference in annual mean wind speed at 10 m or WPD inferred from wind speeds at 10 m is a poor indicator of the

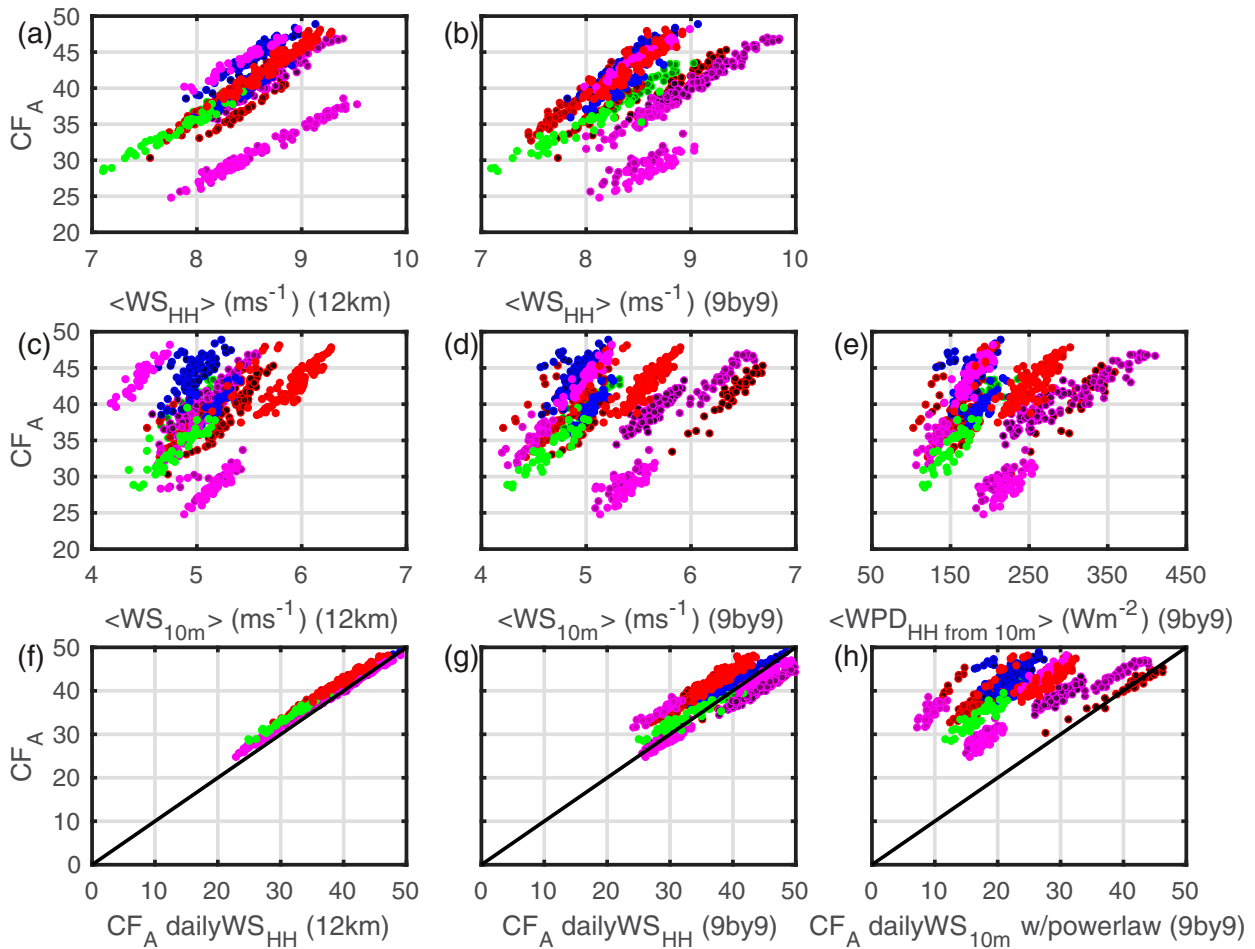


FIG. 4. Scatterplots of annual mean capacity factors at each wind farm derived from hourly hub-height wind speeds (CF_A) vs (a) annual mean hub-height wind speed (WS_{HH}) at 12-km resolution, (b) annual mean wind speed at HH computed from hourly output averaged to 108-km resolution, (c) annual mean wind speed at 10 m AGL (WS_{10m}) at 12-km resolution, (d) spatially averaged annual mean wind speed at 10 m AGL [WS_{10m} (9 by 9)], and (e) annual mean WPD computed from hourly spatially average wind speed at 10 m AGL and corrected to HH using a power law with an exponent of 1/7. Also shown are results pertaining to temporal averaging wherein the value CF_A derived from hourly output near HH (y axis) is plotted against CF_A computed from (f) daily mean wind speeds at HH, (g) daily mean spatially averaged wind speeds at HH, and (h) daily mean spatially averaged wind speeds at 10 m extrapolated to HH using a 1/7 exponent. The wind farm asset locations are colored by the regions shown in Fig. 3, but the interior of the circles is shaded differently for each wind farm.

likelihood of a difference in CF_A , and the response in CF_A to a fractional change in annual mean wind speed at 10 m is location specific. Last, use of daily mean wind speeds leads to a systematic negative bias in CF. For these reasons, although a number of studies have used daily mean wind speed at 10 m that is output from ESM to infer information about possible wind resources and/or wind turbine power production under future climate conditions, we strongly recommend against this and advocate for use of appropriate downscaling.

b. Evaluation of WRF-derived CF

Figure 5 shows the relationship between the modeled mean CF ($\langle CF_{Mod} \rangle$) computed using hourly wind speeds from WRF at hub height and the asset specific power curve versus the mean observationally derived $\langle CF_{Obs} \rangle$ computed from the expected power estimates provided by the wind

farm owner/operator. For the 14 operating locations with more than 3 years of expected power estimates, $\langle CF_{Mod} \rangle$ and $\langle CF_{Obs} \rangle$ are correlated, with Pearson (r) and Spearman (ρ) correlation coefficients that are greater than 0.6. Thus, the WRF simulation at 12-km grid spacing captures *some*, although certainly not all, of the large-scale variability in wind resources across eastern North America. The average difference in $\langle CF \rangle$ between those derived from the WRF Model output and the observed expected power is close to zero, while a small positive bias in the WRF-derived estimates ($<5\%$) is expected because of the influence of wind turbine wakes. Nevertheless, the agreement is indicative of some degree of simulation skill. However, there are some obvious points of discrepancy. For example, at one site in the Midwest the observations indicate a very low $\langle CF_{Obs} \rangle$ while the WRF output suggests a $\langle CF_{Mod} \rangle$ of almost 46%. This is a location

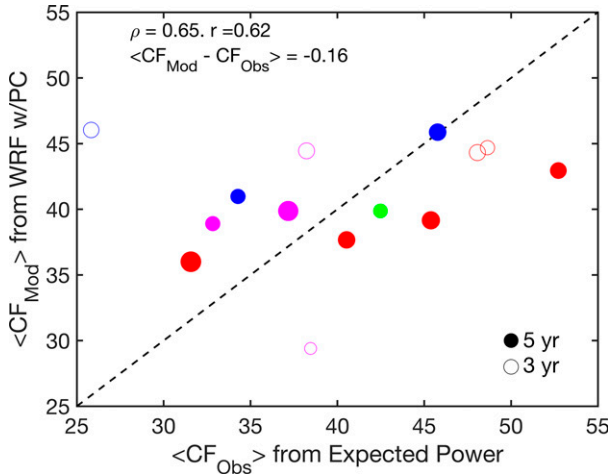


FIG. 5. Scatterplot of mean CF derived from daily expected power ($\langle CF_{Obs} \rangle$) vs those derived from application of the wind turbine power curve to hourly modeled wind speeds (at wind turbine hub height) ($\langle CF_{Mod} \rangle$) for each wind farm that has more than 3 years of observations. Locations with 3 or fewer years of data are shown by the open circles, and those with 5 or more years of data are shown by the filled circles. The size of the circle scales with the installed capacity, and asset locations are color coded by the regions shown in Fig. 3. Numbers in the top-left corner show the Pearson (r) and Spearman (ρ) correlation coefficients and the mean difference between $\langle CF_{Obs} \rangle$ and $\langle CF_{Mod} \rangle$ for the locations with 5 or more years of observations. The dashed line indicates a 1:1 correspondence.

where a scaled wind turbine power curve is applied that might provide an explanation for the poor degree of fit. Further, analyses of expected power from two SGP wind farms that are located within close proximity (<50 km separation) yield $\langle CF_{Obs} \rangle$ of 45.4% and 52.7% (a difference of 7 percentage points) while WRF-derived $\langle CF_{Mod} \rangle$ for these two, nearly adjacent sites, are 39% and 43%, respectively (i.e., a difference of only 4 percentage points).

Many possible explanations for the discrepancy between the WRF-derived CF estimates and the observationally derived CF can be advanced. The most likely three, shown in the rank order of importance as based on the authors' expert judgement, are as follows: 1) The most-likely explanation is differences that are due to interannual variability in the wind regime. As discussed above, any ESM simulation represents a plausible sequence of climate conditions rather than conditions on a specific date. For that reason, the state of critical internal modes as simulated within a specific ESM realization may not be perfectly synchronized with observations. However, their response to external climate forcing due to enhanced greenhouse gas concentrations should be captured [see further detailed discussion in Part II of this series of papers and in Coburn and Pryor (2021)]. Given the importance of different internal climate modes to wind resources in different parts of North America some spatial variability in the degree of agreement is also expected (Pryor et al. 2018). Thus, because the WRF simulations are performed within an ESM there can be no expectation that the output for a given period

represents the actual wind climate in a given year or years. Hence only partial agreement for mean CF computed from 3 to 6 years of simulation versus observations is expected. The postulate that length of record is an important source of error is supported by the scaling of the root-mean-square error (RMSE) between the $\langle CF_{Obs} \rangle$ and $\langle CF_{Mod} \rangle$ across all sites with ≥ 1 year, ≥ 3 years, or ≥ 5 years of complete daily data. RMSE between $\langle CF_{Obs} \rangle$ and $\langle CF_{Mod} \rangle$ for all 23 sites with at least one complete year of daily expected power is 8.2 percentage points. RMSE for the 18 sites with at least 3 years of complete data is 7.6 percentage points and for the 12 sites for 5 or more years of daily CF data it is 5.4 percentage points. This suggests that as the length of record from the observations and model increases both begin to converge on the long-term climatology. 2) The next-most-likely explanation is microscale flow variations that are not captured by the WRF simulations at the current grid spacing of 12 km. Coupling of mesoscale or microscale models or use of high-resolution in the mesoscale simulations generally improves the fidelity of wind resource assessments (Dörenkämper et al. 2020; Haupt et al. 2019), in part due to the improved representation of surface roughness (Dörenkämper et al. 2020; Hahmann et al. 2020) and topographic channeling in regions of complex orography (Horvath et al. 2012; Jiménez and Dudhia 2013). 3) The third-most-likely explanation is that there is uncertainty in the expected power estimates.

To contextualize the RMSE between our WRF simulations and the observed CF, comparisons are also made with output from the NREL Wind Prospector. To help to illustrate the importance of model resolution, these analyses are performed for different displacement distances between the real operating wind farms and the point output from the Wind Prospector. To illustrate the importance of terrain complexity these analyses are performed first using output only from the WP:EWD that reflects locations in relatively low terrain complexity and then for wind farm locations in both the WP:EWD and WP:WWD domains. Recall that both WP:EWD and WP:WWD report estimated CF based on 3 years of WRF simulation at $\Delta x = 2$ km. Comparison of $\langle CF \rangle$ from WP:EWD estimates within a $25 \text{ km} \times 25 \text{ km}$ grid box centered on each operating wind farm, and $\langle CF_{Obs} \rangle$ for sites with ≥ 5 years of expected power yields an RMSE of 3.6 percentage points and a Spearman correlation coefficient ρ of 0.8. This RMSE is lower and ρ is higher than equivalent numbers for current WRF simulation (RMSE = 5.4 percentage points; $\rho = 0.65$), likely due in part to the higher resolution used in the WP:EWD WRF simulations. The RMSE between CF estimated from the 3-yr WRF simulation on which the WP:EWD is based and the first six years of the MPI-WRF simulation at these wind farms is 3.9 percentage points. Thus, CF from these two entirely independent WRF simulations are in relatively close accord. Comparison of $\langle CF \rangle$ from WP:EWD with those from the operating wind farms also illustrates the degradation of model skill with spatial averaging. When WP:EWD output is averaged within a $12.5 \text{ km} \times 12.5 \text{ km}$ grid box around each wind farm site, the RMSE between $\langle CF_{Obs} \rangle$ and those from the WP:EWD is 3.5 percentage points. When the grid box is enlarged to $25 \text{ km} \times 25 \text{ km}$, the RMSE is 3.6 percentage points. When the grid box is enlarged to $50 \text{ km} \times 50 \text{ km}$, the RMSE is

4.7 percentage points. When the analysis is extended to consider operating wind farms covered by the WP:EWD and WP:WWD domains, the RMSE between CF from the Wind Prospector and the observed CF increases to 5.54 percentage points for a grid box of $12.5 \text{ km} \times 12.5 \text{ km}$. Thus, the RMSE between observed mean CF and CF from the Wind Prospector output averaged to 12.5 km and between observed mean CF and those from the current WRF simulations performed using $\Delta x = 12 \text{ km}$ are very similar: 5.54 and 5.36 percentage points, respectively. Recall that we postulate that changes in wind conditions (CF and wind drought/bonus period frequency) due to climate nonstationarity are likely to derive from changes at larger spatial scales and thus should be captured by the MPI-WRF simulations. However, it is acknowledged that any changes in wind direction and their interaction with orographic forcing may not be fully resolved.

Our research methodology uses monthly cold restarts to ensure WRF closely follows the driving ESM, with a 6-h spinup period to preserve computational resources. This could lead to discontinuities in soil moisture (SM). Time series analyses do not indicate SM discontinuities after each restart except over parts of the SGP and Midwest. They also reveal no temporal trend in ΔSM after the restarts over the 40-yr simulation. Nevertheless, SM discontinuities, where present, could impact the surface energy balance and near-surface stability, which may have implications for wind speeds close to wind turbine hub height (Pryor et al. 2020d). Accordingly, Fig. 6a shows the fractional change in soil moisture (ΔSM) over a 6-h period subsequent to cold restarts for a grid cell in the Midwest, along with ΔSM for an equal number of periods displaced from the cold restarts that represent the range of ΔSM that arise purely from atmospheric processes. Whereas the median ΔSM 6 h after cold restarts is less than 1%, at 12 h after the restart nearly 10% indicate ΔSM of greater than 5%. The 90th-percentile ΔSM computed using different time lags Δt indicates convergence between the restart and nonrestart period for lags of 42–48 h (Fig. 6b). These analyses imply that use of a 2-day spinup might be more appropriate for future studies in some locations, but the temporal trends in wind resources reported herein are not likely to be strongly influenced by these relatively short-lived SM discontinuities.

The observational records of CF_{obs} are too short to be used to evaluate IAV, but as described in the introduction, if one assumes the AEP values are Gaussian distributed a frequently applied assumption is that the standard deviation σ of AEP is 6% of the mean (Brower 2012; Pullinger et al. 2017; Raftery et al. 1998). Using nonparametric approaches, a prior WRF study over North America found the IQR(AEP) was ± 3 percentage points around the 50th-percentile AEP (Pryor et al. 2018). CF from the current simulations are consistent with these estimates. At an example, relatively low CF, site in the SGP the median (50th percentile) CF_A over the 40 years of simulation is 36.7% and the IQR(CF_A) extends from 35.8% to 38.9% (i.e., IQR = 3.1 percentage points) (Fig. 7). Estimates for all wind farm locations indicate that the IQR varies around the median CF_A (i.e., long-term average over the 40 years) within 7.5 percentage points. That is, the middle half of CF_A values lie within ± 3.8 -percentage points of the median

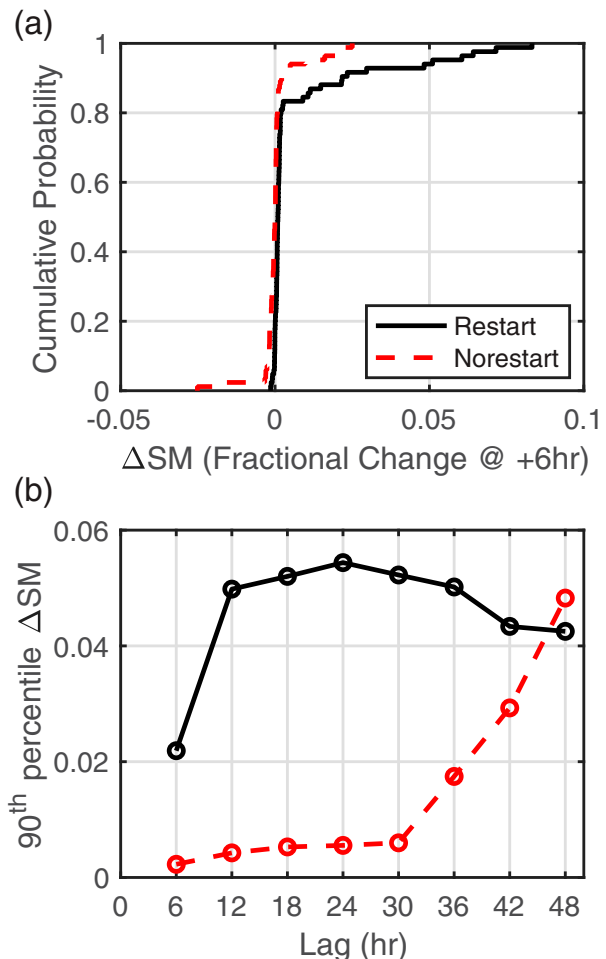


FIG. 6. (a) Cumulative probability plots of fractional change in soil moisture in the 6 h after each monthly cold restart for an illustrative grid cell in the Midwest. Also shown is ΔSM over an equivalent number of periods without a restart (Norestart). (b) The 90th-percentile value of ΔSM in the Restart and Norestart samples for varying lags (i.e., hours since the Restart or Norestart) for the same grid cell as shown in (a).

CF_A at that location. The spatially averaged ratio of the $\sigma(\text{CF}_A)$ to $\langle \text{CF}_A \rangle$ is 0.054 [i.e., $\text{IAV}(\text{CF}_A) = 0.054$]. The $\text{IAV}(\text{CF}_A)$ estimate for the SGP (7 locations) is 0.057, while for the Northeast (7 locations) it is 0.050.

c. Evaluation of CF seasonality and probability distributions

Recall that the MPI-ESM-LR are free-running simulations and as such the modeled time series from WRF will not match the observations directly, but both the seasonal cycle and the cumulative probability distribution of daily CF values should match the observations. A comparison of the 30-day running-mean CF from observations and the first five years of model-derived values for illustrative wind farms in the southern Great Plains and the Northeast is summarized in Fig. 8. The postprocessed WRF Model wind speeds result in 30-day

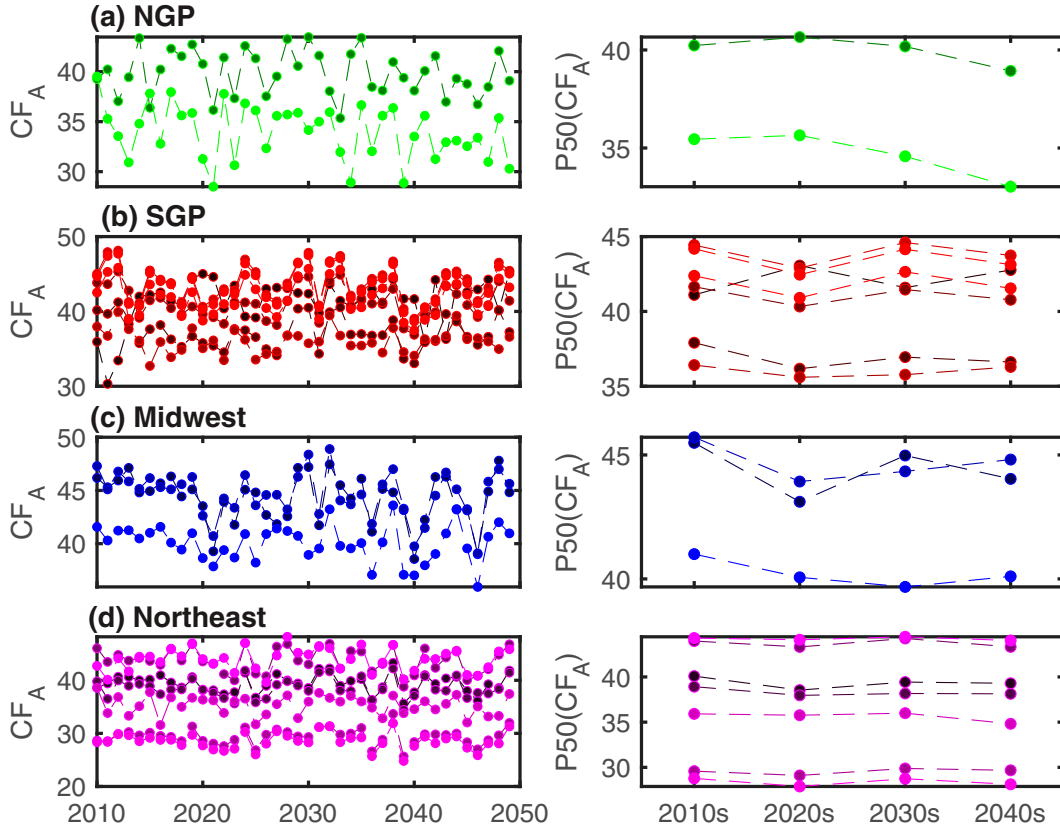


FIG. 7. (left) Time series of the annual mean CF computed by postprocessing hourly wind speeds from WRF at the wind turbine hub height using a location-specific power curve, and (right) the 50th-percentile CF_A in each decade. The regions are listed in west–east order: (a) northern Great Plains, (b) southern Great Plains, (c) Midwest, and (e) Northeast. Wind farm locations are color coded by the regions shown in Fig. 3, with different locations in a given region denoted by the varying hues (as in Fig. 4).

running-mean daily CF at this wind farm, as with other locations in the SGP, that capture some features present in the observations. For example, the model captures the relatively low production in the summer (low 30-day running-mean CF) (Fig. 8). However, the recovery to higher production in the fall is delayed relative to the observations. There is good consistency with the observations in terms of the highest 20% of 30-day running-mean CF. The model-derived 80th-percentile 30-day running-mean CF at this location is 54.5%, whereas the equivalent value from the observations is 55%. However, the lowest 20% of running 30-day mean CF from the model are biased low. The estimated CF associated with the lowest 20% of running 30-day mean values is 35.2% in the model output and 41.2% in observations. Thus, the WRF simulations are generating more frequent, and more intense, extended periods of lower production in the summer than are manifest in the observations. This may be due to too high a persistence in the MPI ESM that provides the LBC or the imperfect representation of mesoscale/thermo-topographic features in the WRF Model when applied at 12-km grid spacing. At the example site in the Northeast, and the majority of other sites in this region, the degree of agreement in terms of the seasonality of 30-day running-mean CF and the probability distribution of daily CF between the model and the observations is

better (Fig. 8). However, the summertime minimum is displaced in time and model-derived CF remains too high in the early summer relative to the observations. Hence, the lowest 20% of 30-day running-mean CF are positively biased in the postprocessed model output. The model-derived value at this location is 29.4%, while the equivalent value from the observations is 23.4%. There is closer accord for the highest 20% of running 30-day mean CF. The estimated CF associated with the highest 20% of running 30-day mean values is 54.5% from the model output and 55% in observations. We are not aware of any previous attempt to evaluate the ability of numerical models to simulate the magnitude of extended wind droughts and/or wind bonus periods, so it is not possible to assess the relative credibility of these simulations. The cumulative probability distributions of daily CF for the two example sites (Figs. 8b,d) show good accord with those derived from observations.

d. Projections of capacity factors

CF_A time series from the 40 years of WRF simulations are subject to Mann–Kendall tests to evaluate the presence of monotonic trends. The results indicate that at 26 of the 27 locations the CF_A values exhibit no significant trend. Only one of the two sites in the NGP exhibits a significant negative trend in

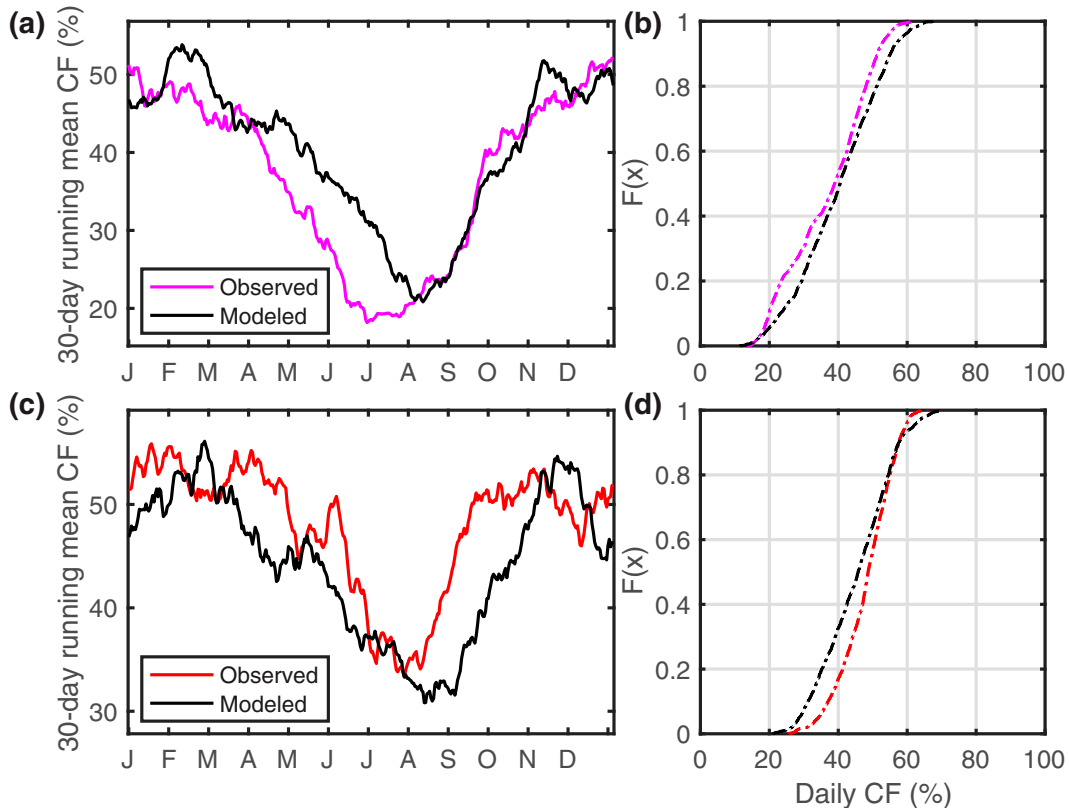


FIG. 8. (left) Day-of-year mean 30-day running-mean CF from the postprocessed model output and the observations and (right) cumulative probability distributions of daily CF from the postprocessed model output and the observations at an example wind farm located in the (a),(b) Northeast and (c),(d) SGP.

CF_A at the 95% confidence level. Rather, the CF_A time series from all locations appear to be dominated by continued inter-annual variability (Fig. 7) likely linked to the action of internal climate modes and resulting low-frequency variability.

When the 50th percentile CF_A values in each decade (2010–19, 2020–29, 2030–39, 2040–49) are computed at the wind farms, $P50(CF_A)$ at locations in the SGP do not show a consistent tendency toward either increased or decreased values (Fig. 7b). The mean difference (averaged across all seven sites) in $P50(CF_A)$ in the 2040s minus the 2010s is -0.4 percentage points. At some wind farms the $P50(CF_A)$ in the 2040s is higher than in the 2010s. However, the $P50(CF_A)$ in the 2040–49 simulation period does not lie beyond the 95% confidence intervals computed for 2010–19 at any of the SGP wind farms. Modeled $P50(CF_A)$ for the three wind farms in the Midwest exhibit evidence of increased IAV(CF_A) in the 2030s and 2040s (Fig. 7c). All three have a $P50(CF_A)$ that is significantly lower in the 2040–49 period (mean difference of -1 percentage points). The two wind farms in the NGP show some evidence of slightly lower $P50(CF_A)$ (by up to 2 percentage points) in the 2030s and 2040s (Fig. 7a). Only one has a $P50(CF_A)$ in 2040–49 that lies beyond the 95% confidence interval from 2010 to 2019. No change in the IAV(CF_A) or decadal estimates of $P50(CF_A)$ are found for six of the

seven wind farms in the Northeast (Fig. 7d). The mean difference (averaged across all seven sites) in $P50(CF_A)$ in the 2040s minus the 2010s is -0.6 percentage points, with some sites showing higher values in the 2040s, and others showing higher values in the 2010s.

e. Projections of “wind droughts” and “wind bonus” periods

Based on the preliminary assessment of the cumulative density functions of daily CF and the seasonality of 30-day running-mean CF from the WRF output summarized above, first estimates of the probability of wind droughts and bonus periods in each decade of the simulation at each wind farm are computed (Fig. 9). The results indicate that the sites within a given region tend to behave in a coherent manner. For example, decades with a higher probability of wind bonus periods are manifest at most/all sites in that region (e.g., the 2010s in the SGP). This is consistent with the concept that extended periods of anomalously high or low wind power production are derived from large-scale climate dynamics. With the exception of the two wind farms in the NGP, results for the last decade of the analysis (the 2040s) all lie within the range of the previous three decades. Thus, for this model chain, there is no clear emergence of a signal of increased or

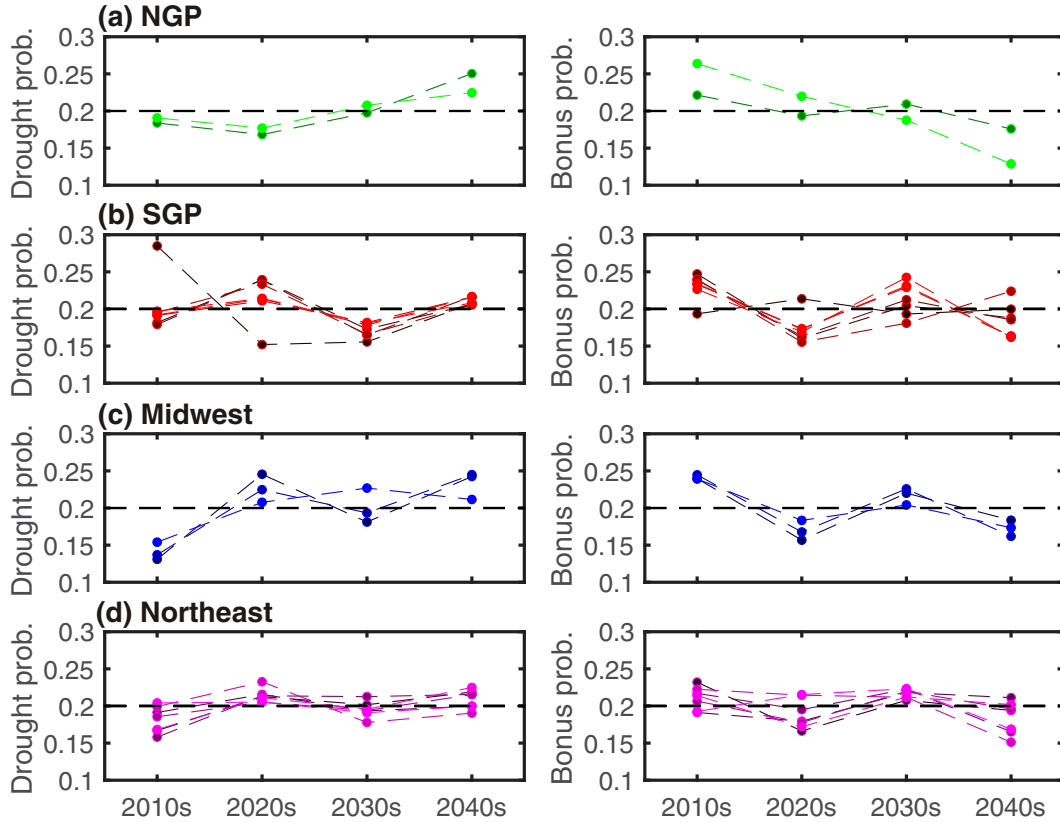


FIG. 9. Time series of the probability that any 30-day period in the given decade (left) will be a wind drought (i.e., will fall below the 20th-percentile 30-day running-mean CF for that period of the year) or (right) will be a wind bonus period (i.e., exhibit anomalous high CF that fall above the 80th-percentile 30-day running-mean CF for that period of the year). The regions are listed in west–east order: (a) northern Great Plains, (b) southern Great Plains, (c) Midwest, and (e) Northeast. Wind farm locations are color coded by the regions shown in Fig. 3, with different locations in a given region denoted by the varying hues (as in Fig. 4). The horizontal dashed black line in each panel shows a probability of 0.2.

decreased probability of prolonged wind droughts or wind bonus periods. At the two wind farms in the NGP, this simulation implies there is a higher probability (by 2–4 percentage points) of an extended period of anomalously low electricity production in the 2040s, a decade that also exhibits a lower probability (by 3–8 percentage points) of a wind production bonus in this region.

4. Summary and concluding remarks

Although use of dynamical and/or statistical downscaling to make projections of wind resource and power production under climate nonstationarity involves penalties in terms of computational and personnel cost, as illustrated herein (see Fig. 4), use of direct output from ESM is not justifiable. There is clear enhancement in terms of utility to the wind energy industry from enhanced model resolution, and storage of high-frequency (hourly) wind speeds at/near wind turbine hub heights.

Once-hourly wind speed outputs at wind turbine hub height from WRF simulations are postprocessed using location-

specific wind turbine power curves. The resulting mean CFs from the postprocessed WRF output are shown to capture some of the macroscale in observed CF derived from expected power at operating wind farms (Fig. 5). This enhances confidence in projections of future power production.

Projections to the middle of the twenty-first century, for a high climate forcing scenario, indicate stability of annual mean CF (CF_A) to 2050 (Fig. 7), continuing variability consistent with the action of key internal climate modes and no secular trend. $P50(CF_A)$ values projected for 2040–49 lie within the 95% confidence intervals from 2010 to 2019 at 14 of the 19 wind farms considered herein. It is noteworthy, however, that the five wind farm locations that do exhibit significant differences all indicate lower $P50(CF_A)$ in the future. Three are located in the Midwest, one in the NGP and one in the Northeast. When the 50th percentile CF_A values in each decade (2010–19, 2020–29, 2030–39, 2040–49) are computed at all wind farms, $P50(CF_A)$ at locations in the southern Great Plains (SGP) do not show a consistent tendency toward either increased or decreased values (Fig. 7b). The mean difference, averaged across all seven SGP sites, in $P50(CF_A)$ in the 2040s

minus the 2010s is -0.4 percentage points. At some SGP wind farms the $P50(CF_A)$ in the 2040s is higher than in the 2010s and at no site does the $P50(CF_A)$ in the 2040–49 simulation period lie beyond the 95% confidence intervals computed for 2010–19. There is evidence for declining CF_A in the northern Great Plains by the 2040s, with the $P50(CF_A)$ lower than those in the 2010s by up to 2 percentage points (Fig. 7). This possible reduction in $P50(CF_A)$ at the wind farms in the NGP is smaller than the current $I\Delta V(CF_A)$ and thus is within the envelope of current experience. It is also important to note that only one of the NGP locations had an observational record of > 3 years duration and thus limited fidelity assessment can be performed. For the site with ≥ 5 years of expected power estimates the mean CF ($\langle CF \rangle$) derived from the WRF Model output is biased low by 2.6 percentage points (Fig. 5). The $\langle CF \rangle$ estimate from expected power is 42.5%, while that from the postprocessed WRF Model is 39.9%.

While no previous research has generated equivalent projections of CF_A over subregions of eastern North America, past research has also indicated relative stability of the wind climate to the midcentury, with small magnitude declines over much of the eastern United States and increases or stable wind resource in the SGP (Fig. 2). Most of these analyses have used time-slice ESM-RCM experiments where the regional simulations are performed for specific time windows rather than as a continuous time series as here. Nevertheless, the results show broad consistency in terms of both the sign and magnitude of differences. For example, statistical analyses based on one global–regional coupling indicated that the annual mean wind speed close to typical wind turbine hub height in 2040–69 was within $\pm 10\%$ of contemporary climate (1985–2005) in all seasons (Haupt et al. 2016). An analysis of 90th-percentile wind speeds at 10 m AGL based on statistical downscaling of the North American Climate Change Assessment Program (NARCCAP) regional models (grid spacing of 50 km) found evidence for a 3% decrease over most of the eastern United States in 2041–60 relative to 1981–98 (Pryor and Barthelmie 2014). An additional independent analysis based on wind speed output at 10-m height from the NARCCAP regional models suggested the WPD in 2038–70 was higher than in 1968–2000 across much of the southern Great Plains (in Kansas, Oklahoma, and northern Texas) (Johnson and Erhardt 2016). An analysis of WPD based on regional model output wind speeds at 10 m AGL indicated higher values over Kansas (a 2%–3% increase) in 2040–70 versus 1970–2000 but declines over central Colorado of up to 1%–2% (Greene et al. 2012). Independent WRF simulations (grid spacing of 12 km) with the MPI-LR realization used here but a longer time horizon (and the RCP8.5) found evidence for increased AEP (by up to 8%) in 2075–99 versus 1980–2005 in the southern Great Plains but declines of up to 5% in the northern Great Plains (Pryor et al. 2020c). An analysis of WPD based on daily 10-m wind speeds vertically interpolated to HH using the power law (with a $1/7$ exponent) from the North American Coordinated Regional Climate Downscaling Experiment (NA-CORDEX) regional simulations (eight global and five regional models applied at either 25- or 50-km grid spacing) found a consistent signal of higher values over the SGP in 2071–2100 relative to 1971–2000,

particularly during the summer, and small declines over the NGP (Chen 2020). The multimodel ensemble mean differences in those regions (2071–2100 minus 1971–2000) are $+8\%$ and -4% to -6% , respectively (Chen 2020).

A new approach is developed and applied to the WRF output here to assess possible changes in the probability of extended wind droughts and bonus periods. These analyses are subject to comparatively large uncertainty due to the relatively short record used to make the climate normal against which each 30-day running mean is compared. Nevertheless, initial evidence suggests the postprocessed WRF output reproduces the probability distribution of daily CF and implies some ability to represent key aspects of observed “wind droughts” and “wind bonus” periods in the contemporary climate. This enhances the confidence with which projections can be viewed. Seasonally normalized extended periods of anomalously high (bonus) and anomalously low (drought) CF do not exhibit evidence of changing frequency over most of the eastern two-thirds of North America. Locations in the NGP exhibit a greater frequency of extended wind droughts by the midcentury, and a decline in the likelihood of “wind bonus” periods (Fig. 9).

It is useful to consider the possible changes in $P50(CF_A)$ from a financial perspective. There is tremendous variability in the price paid by electricity consumers and producers but according to the U.S. EIA in 2020 the average retail electricity price ~ 10.59 cents per kW h (<https://www.eia.gov/energyexplained/electricity/how-electricity-is-generated.php>) and the average wholesale price was 3.39 cents per kW h (<https://www.eia.gov/electricity/wholesale/#history>). Assuming this wholesale price is uniformly applicable, for a wind farm with the mean installed capacity of those considered here of 170 MW, assuming a capacity factor of 40% in the contemporary climate, the annual revenue is $\sim \$20$ million, and thus a 1-percentage-point increase or decrease equates to a \$200,000 change in revenue. Many wind farms operating in North America signed 20-yr fixed price power purchase agreements (Miller et al. 2017) several years ago that may shortly end, making the accuracy of resource assessments more important. Scaling across the entire industry with current IC of 135 GW, there is clear value in continuing efforts to make more robust projections of wind climate variability and change to select optimal locations for future wind energy installations.

It is also useful to contextualize the projected differences in CF_A in historical gains in CF from operational wind farms due to technological advances (e.g., increased wind turbine reliability). A recent analysis suggested the “average capacity factor in 2020 exceeded 40% among wind projects built in recent years and reached 36% on a fleet-wide basis. The average 2020 capacity factor among projects built from 2014 to 2019 was 41.4%, as compared with an average of 29.0% among projects built from 2004 to 2011, and 25.2% among projects built from 1998 to 2001” (Wiser et al. 2021). Thus, technology-driven changes in historical CF equate to an average annualized increase of approximately 0.75 percentage points per year. These historical changes in CF greatly exceed projected changes in CF_A due to global climate nonstationarity.

The computational demands and data storage requirements associated with performing long-term simulations at 12-km

resolution over such a large domain limit this analysis to use of a single model chain: WRF nested within a single realization, for a single climate forcing from a single ESM (MPI-LR). This is thus a key source of uncertainty in the projections presented here. The second part of this combined dynamic–statistical downscaling explores more of the uncertainty space associated with the climate forcing and ESM by adopting a highly computationally efficient statistical downscaling approach to develop a suite of annual CF projections. That analysis is designed to explore the degree to which projected changes in CF and wind droughts and bonus periods are consistent across ESM and climate forcing scenario, and to provide a first assessment of to what extent, and when, externally forced trends (i.e., signals) will emerge relative to internal variability (i.e., noise) in the climate system (Hawkins and Sutton 2009; Hawkins et al. 2020).

A key bottleneck in making credible projections of wind power generation is the paucity of publicly available datasets of wind power production (Kuskiak 2016) with which to evaluate numerical model output in the contemporary climate and to train statistical downscaling models. This challenge was also recognized in the NEWA project that instead used an evaluation methodology of the contemporary climate based on nonconcurrent measurements of mean wind speed on tall towers (Dörenkämper et al. 2020). Therefore, although the current study is a unique contribution to the literature, it must be emphasized that here we consider 27 operating wind farms and use the consistency across the subregions to examine the presence/absence of consistent trends within those regions. Future work should seek to expand to consider additional operating wind farms and work with owner-operators to make further power production estimates available for industry-relevant climate research.

It is also important to acknowledge that regional simulations performed with a 12-km grid spacing, such as those presented herein, do not fully capture all of the dynamic forcing of wind resources and electricity generation potential. Here we implicitly assume that the climate change signal, if any, is carried by larger scales of atmospheric motion that are comparatively well represented by the MPI-WRF Model chain. Other sources of uncertainty associated with the dynamically downscaled projections presented herein pertain to selection of the physics schemes. Biases in wind speeds at/close to typical wind turbine hub heights exhibit a clear dependence on factors such as the PBL scheme applied, source of the land-use/land-cover data, and domain size, in addition to the source of the lateral boundary conditions (Hahmann et al. 2020). Future work will consider whether use of other surface and/or PBL schemes that better account for land use/terrain variability and use of higher grid spacing in the long-term WRF simulation alter the sign or magnitude or spatial manifestations of the climate change signal in wind resources and/or drought and bonus periods.

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Data availability statement. The wind farm proprietary data are covered by a nondisclosure agreement and cannot be distributed. The WRF Model output is available upon request from the authors. CMIP5 data including that for the MPI-LR-ESM are available online (<https://pcmdi.llnl.gov/index.html>), as are ERA5 data (<https://climate.copernicus.eu/climate-reanalysis>; <https://doi.org/10.24381/cds.adbb2d47>). Wind Prospector data are also available online (<https://maps.nrel.gov/?da=wind-prospector>).

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